

Week 7 Testing Report — Generative AI Track

Every Week 7 test documented as **Test** → **Finding** → **Fix** → **Retest** → **Evidence**.

T1. Booking flow crashes on insert

Field	Value
Component	AppointmentService
Scenario	Chat booking → <code>create_appointment</code>
Expected	Appointment row inserted, APT-XXXX returned
Actual	<code>AttributeError: 'str' object has no attribute 'hex'</code>
Issue	<code>patient_id</code> passed as Python <code>str</code> to a SQLAlchemy <code>UUID(as_uuid=True)</code> column
Fix	Coerce with <code>uuid.UUID(patient_id)</code> before constructing the ORM row
Retest	APT-3EE571AA created and visible in DB

T2. New conversation created on every turn

Field	Value
Component	ChatService
Scenario	15-turn conversation with a single <code>session_token</code>
Expected	One conversation row, history persists
Actual	5 separate conversation rows for 15 turns — memory never fired
Issue	<code>_create_conversation</code> inserted without checking existing rows by <code>session_token</code>

Fix	Idempotent creation: lookup by <code>session_token</code> first, reuse or insert
Retest	Single row after 15 turns; rolling summary and facts persisted

T3. Memory block ignored by LLM

Field	Value
Component	ConversationAgent system prompt
Scenario	"What is my name and what am I allergic to?" after prior facts
Expected	"Your name is David and you're allergic to aspirin"
Actual	KB chunk echoed instead
Issue	System prompt forbade using non-KB content
Fix	Rule 5 added: authorize prior-conversation block as authoritative for user facts
Retest	Correct answer returned across multiple runs

T4. Session_token UNIQUE constraint violation

Field	Value
Component	ChatService <code>_ensure_conversation_db</code>
Scenario	SSE request triggering two writes with the same <code>session_token</code>
Expected	Reuse existing conversation
Actual	<code>sqlite3.IntegrityError: UNIQUE constraint failed: conversations.session_token</code>
Fix	Lookup by <code>session_token</code> before insert in <code>_ensure_conversation_db</code>
Retest	No more UNIQUE errors, SSE requests succeed

T5. Escalation not triggering on common phrasings

Field	Value
Component	ChatService booking/escalation short-circuit
Scenario	"connect me" after out-of-scope question
Expected	<code>requires_human=True</code>
Actual	Polite repeat of the same offer
Issue	Escalation intent set was narrow; didn't include conversational phrasings
Fix	Expanded trigger set to 17+ phrasings
Retest	"connect me" and "why arent you connecting me" both return <code>requires_human=True</code>

T6. `_pending_escalations` NameError

Field	Value
Component	ChatService <code>process_message</code>
Scenario	Turn 3 of escalation test
Expected	Continue escalation
Actual	<code>NameError: name 'pending' is not defined</code>
Issue	Partial patch replaced one reference, left another
Fix	Module-level <code>_PENDING_ESCALATIONS</code> dict; all references normalized
Retest	6-turn escalation test: 5/6 correct (final fix in T7)

T7. Pending escalation cleared too eagerly

Field	Value
Component	ChatService escalation short-circuit
Scenario	"yes" after escalation
Expected	Confirm escalation again
Actual	LLM asked "Would you like me to connect you?"
Issue	Flag popped immediately after escalation
Fix	Keep flag set; auto-expire only on bare confirmation (yes/ok/sure)
Retest	6-turn test: 6/6 correct

T8. Weekday not parsed for booking

Field	Value
Component	BookingService _rule_extract
Scenario	"saturday at 11am"
Expected	Resolve to next Saturday's ISO date
Actual	No date extracted
Fix	Added weekday resolver mapping to next occurrence
Retest	date=2026-09-19, time=11:00 extracted

T9. SSE streaming returns 500

Field	Value
Component	/chat/message/stream endpoint
Scenario	Any streaming query

Expected	SSE frames stream to browser
Actual	500 — <code>AttributeError: 'NoneType' object has no attribute 'get'</code>
Issue	<code>retrieval_result.output</code> was <code>None</code> when 0 chunks retrieved
Fix	<code>Guard_out(r) = r.output or {}</code> in <code>orchestrator._assemble_response</code>
Retest	SSE returns 200 with <code>text/event-stream</code> , frames stream

T10. Frontend streaming never reached the browser

Field	Value
Component	<code>src/lib/api.ts</code> → <code>sendMessageStream</code>
Scenario	Chat streaming UI
Expected	Chunks render progressively
Actual	Browser hit a Next.js rewrite URL that buffered SSE
Fix	Frontend calls backend directly: <code>\${NEXT_PUBLIC_API_URL}/chat/message/stream</code>
Retest	<code>curl</code> , browser console, and app all see chunk-by-chunk output

T11. Response duplication in chat UI

Field	Value
Component	Zustand <code>chat-store.addMessage</code>
Scenario	SSE chunks arriving one at a time
Expected	Single bubble that updates

Actual	Every chunk appended → bubble repeated 8×
Fix	<code>addMessage</code> replaces by <code>id</code> if already present
Retest	Single bubble types in progressively; no React key warning

T12. Terminal mojibake "€"

Field	Value
Component	Windows Git Bash display
Scenario	Reading chat responses in terminal
Expected	Clean Unicode em dashes and narrow spaces
Actual	"", "€" shown
Investigation	Python-native response capture shows U+2013, U+202F — correct
Conclusion	NON-ISSUE: terminal display artifact, not API bug
Retest	Browser shows clean Unicode

Summary

Status	Count
Total tests	12
Real issues fixed	10
Non-issues verified	1
Documentation fixes	1

Full raw evidence in `evidence/`.